Instr Sci (2012) 40:755–768
DOI 10.1007/s11251-012-9211-z

What and how advanced medical students learn from reasoning through multiple cases

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Received: 14 February 2012 / Accepted: 15 March 2012 / Published online: 21 April 2012
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Abstract The study reported in this article concerns the questions what and how fourth-year medical students can learn from a series of cases that have a similar underlying problem. This question is crucial in the theoretical sense as it looks at mechanisms of updating and improving knowledge structures, which are conjectured to consist of 'illness scripts', which integrate knowledge about enabling conditions for a disease, the underlying fault, and the consequences or signs and symptoms of the disease. It was found that much learning was implicit, whilst students explicitly paid attention to variations in enabling conditions and consequences, which led to improvement of these knowledge components. Integration of biomedical knowledge into the fault component was, however, hampered by lack of knowledge and by misconceptions that students were unable to debug in the absence of learning resources or detailed feedback. The practical importance of this study

Electronic supplementary material The online version of this article (doi:10.1007/s11251-012-9211-z) contains supplementary material, which is available to authorized users.

To be included in Instructional Science, special issue: Collaborating with digital tools and peers in medical education. Cases and simulations as interventions in learning. Editors: L. Helle and R. Säljö

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stems from its close parallels to the learning situation in clinical practice, where patients (or cases) are the most important learning resources, whilst feedback on results is often minimal. Implications and possible solutions for clinical teaching are discussed.

Keywords Learning from cases · Knowledge restructuring · Knowledge integration · Illness scripts · Misconceptions

Learning medicine

One of the things you learn at a very young age is that certain conditions in your environment or in your own behaviour can lead to pain and other discomfort. This knowledge is learned from lessons of the format <do/don't do this—or else—you'll get ill>. Sometimes these lessons are reinforced by the experience warned for, which in turn can then be strengthened by the final part of the lesson <I told you!>. The following are examples of such lessons: "Don't eat raw meat, or else you'll get food poisoning"; "put on your coat, or else you'll catch a cold" and "wash your hands, or else you'll catch the flu." In answer to children's ceaseless questions 'Why?' and 'What is that?', these lessons can soon become rather complex. The first author remembers that her aunt once forbade her to lick the bowl clean in which this aunt had prepared cake dough, a treat otherwise, but now it had been made with duck eggs, two large, greenish eggs with big, very yellow yolks. Her question 'Why?' received the answer that it might give her typhoid fever, a disease—she was told—that had almost killed her mother when she was a young girl.

Such lessons result in knowledge structures that link enabling conditions (entities, events, situations that can threaten your health) to consequences (pain, fever, and worse), and the behaviours needed for prevention or cure (take medicine, get a bandage, bed rest, chicken soup, etc.). These elements are held together with stories about 'How' that can happen, and that teach children about the faults, which are the underlying mechanisms of genetic conditions, germs, defence, healing, etc. These knowledge structures can be seen as the rudimentary forms of more generic scripts that develop with experience, with slots that have default values and can be instantiated with the specific value of a certain case, internally linked by a temporal and causal structure (Schank and Abelson 1977). The scripts have been called 'illness scripts' by Feltovich and Barrows (1984). Illness scripts consist of enabling conditions for a certain disease, the consequences of that disease, and the biomedical fault that explains how the enabling conditions can lead to consequences. Physicians' clinical experience has resulted in very extended, detailed, and differentiated illness scripts. The scripts are organised in memory based on their similarities and differences, and on levels of abstraction and generality. Activated and instantiated illness scripts generate a diagnosis if the case findings have acceptable slot values. Otherwise, related scripts are activated or a temporary mental model is created, in which the fault mechanism is provisionally adapted based on other knowledge and case findings (Feltovich and Barrows 1984; Lesgold et al. 1988).

The study of medicine is very much the study of faults, especially in the first years of traditional curricula when the biomedical topics are taught. Later the clinical courses follow that describe the appearance, epidemiology, pathophysiology, course, treatment, and management of diseases. In practice, both kinds of knowledge are needed, which requires that they be integrated. How important biomedical knowledge is in practice and when this integration takes place have been much debated in the medical education literature. Many practitioners stated and still state that the medical curriculum was

overloaded with biomedical knowledge and that it was hardly used in practice (see, for instance, Bordage 1987). Patel et al. (Patel et al. 1989) argued that biomedical knowledge and clinical knowledge were two worlds apart. Clinical knowledge would be used in clinical reasoning, whilst biomedical knowledge would be used in case explanation. Still others (Schmidt and Boshuizen 1993) claimed that with higher levels of expertise biomedical knowledge and clinical knowledge would become integrated, and that biomedical knowledge was used when needed, in diagnosis as well as in explanation. Reorganisation of biomedical knowledge is needed to integrate it in the fault part of illness scripts.

Knowledge restructuring

Two important findings provided the grounds for the conclusion that higher levels of expertise are associated with higher levels of integration (Schmidt and Boshuizen 1993): First, novices and experts used hardly any biomedical knowledge in diagnosis and case explanation, whilst intermediates did so to a large extent. However, comparison of the case explanations by students and experts with a model explanation revealed that the percentage of concepts that were identical increased sharply with higher expertise levels. Furthermore, a content analysis of the experts' explanation revealed that the concepts used were at an intermediary level of detail, between basic biomedical knowledge and clinical concepts, with a high explanatory power regarding the case findings. Examples are 'sepsis', 'micro-emboli', and 'aortic insufficiency'. Novices and intermediates on the other hand used very detailed concepts, for instance explaining the specific temperature pattern included in the expert concept 'sepsis'. Second, students' performance was largely affected by reducing the case processing time in the experiment, whilst experienced physicians were hardly influenced. Students needed the whole 3 min and 30 s to reason through the case step-by-step activating their detail-level knowledge, and failed this completely when they were given only 30 s. Experts were hardly affected.

Based on these findings and on what is known about the way medical students were educated, Schmidt and Boshuizen (1993) theorised that initially students build networks of thousands of concepts. Through repeated use in solving clinical cases, these networks become validated and connected whilst links are strengthened, and the prominence of certain concepts and groups of concepts changes. In this process, certain higher-level concepts, which Schmidt and Boshuizen called 'encapsulating' concepts, such as the ones given above, come to stand for the entire detailed networks of clinical and biomedical concepts.

This assumption that knowledge restructuring is vital in medical expertise development is further supported by the findings of Woods and colleagues (Woods 2007; Woods et al. 2005), who showed that this kind of integrated biomedical knowledge made clinical knowledge more robust and improved clinical reasoning, whilst Rikers et al. (2005) showed that experts' biomedical knowledge is always co-activated with clinical knowledge even in routine situations. Finally, researchers have found that problem-oriented knowledge structures had a positive effect on diagnostic outcomes compared with knowledge organisation according to generic principles (Coderre et al. 2009; Mclaughlin et al. 2010). Together these studies suggest that continued and repeated experience with classes of similar cases may result in the integration of biomedical knowledge in clinical knowledge as fault knowledge in illness scripts, which in turn has a positive outcome on diagnostic problem solving. Figure 1 demonstrates how biomedical knowledge and clinical knowledge are linked in an illness script.

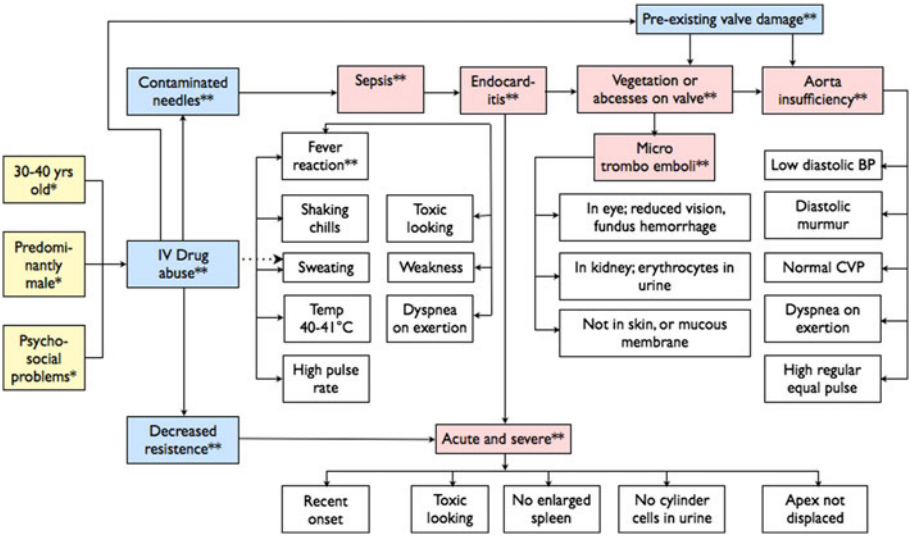

Fig. 1 Example of an illness script of acute bacterial endocarditis for the patient category IV-drug users. *Note*: * Concepts that represent enabling conditions (yellow), ** concepts that represent the fault (blue and pink), other concepts represent the consequences of the fault, i.e., the signs and symptoms of the disease. Note that some fault concepts have a more biomedical flavour than others; e.g., aorta insufficiency or vegetation on valve. Some are closer to the perceptual and clinical level of the signs and symptoms; e.g., fever reaction. (Color figure online)

Implications for instruction and research question

In medical education, the development of such problem-oriented knowledge structures is an important reason for teaching with clinical cases, as is done in problem-based curricula (Boshuizen 2009). However, most curricula struggle with time available to cover so many diseases and basic science topics that no time is spent on the study of sets of similar or related cases to explore variations in the different elements of the illness scripts. Therefore, the further development and differentiation of their illness scripts are left to the epidemiological characteristics of the practices students will participate in later on in their internships.

We consider the underlying assumption questionable that further illness script development will take place automatically once students enter their internships. One reason is that Prince et al. (2000) showed that, at least in the initial stages of learning in practice, energy and attention go to very different practical aspects of practice. Furthermore, no empirical data about the spontaneous learning processes when students meet multiple cases with similar faults exist; hence, we do not have knowledge either about which learning processes can be left to the students' own devices and which processes may benefit from more focussed educational measures. The present study aims to fill in blanks regarding elements and processes of learning from multiple cases and its outcomes. Therefore, we asked medical students who are about to enter their internships to reason through a set of cases with the same underlying problem, gathering information about their reasoning and knowledge development with think-aloud protocols, case explanations, and learner reports.

Method

Participants

Thirty-one 4th-year undergraduate medical students of a medical school with a problem-based learning (PBL) curriculum participated. Data were collected at the end of the academic year, which means that the students had almost completed the preclinical part of the curriculum and were about to enter their internships. In medical school, the students had worked with clinical cases in different settings: paper cases in PBL sessions, simulated patients in the skills lab, and early experience with real patients in the community and in hospitals. The participants were randomly assigned to two groups who received different sets of endocrinological problems. All students repeatedly studied the basic mechanisms and clinical consequences of the endocrinological problems in prior training. Due to no shows and incomplete data sets, one group consisted of 14 students and the other of 17 students.

Materials

We developed two series of five cases with the same underlying pathophysiological process but with different presentations and causes. One series consisted of polycystic ovaries (PCO) cases and the other series of hyperprolactinaemia (HPA) cases. The core of both disorders is an endocrine dysfunction in women leading to menstrual and fertility problems. A study by Dawson-Saunders et al. (1990) with expert clinical and biomedical teachers had earlier indicated that endocrine feedback systems are very important (top 3 of 27) but not difficult (easiest 3 of 27). A practice case and three filler cases concerning other menstrual problems were developed as well. All cases were based on real patients and written in cooperation with a gynaecologist and an endocrinologist. The cases were comparable in length and presented in a standard medical order successively describing history taking, physical examination, lab findings, and additional examinations.

Procedure

The students were told that they would be presented with a series of cases and that their task was to diagnose and explain each case and to learn as much as they could from all these cases. No suggestions were given as to how the students could approach their learning tasks. Cases were presented on a computer screen, and were segmented into meaningful sections consisting of a couple of sentences that represented standard units of history taking, physical examination, lab findings, and additional examinations. A mouse click brought on the next unit. Students processed the cases whilst thinking aloud. The instruction was to express everything that came up in their mind whilst going through a case. Consecutively, the participants gave their diagnoses by typing them directly into a window in the presentation programme and provided a pathophysiological explanation on paper. This format was chosen to enable the use of graphical representations and reflection during explanation. After the students had completed a case, the official diagnosis was given, as they would have received similar feedback if they had encountered the patient as clerks in the hospital.

After the instructions, a practice case was used to make the students comfortable with the procedure. The three filler cases were interspersed as numbers 3, 5, and 7 to ensure attentive case processing.

Finally, after all cases had been processed, the students were reminded of their task to learn from the cases and to tell what they knew now about PCO or HPA (learner reports). The students were probed with the concepts that are neutral to illness script categories, generally used as headers in medical textbooks, and well known to the students: Definition, epidemiology, pathophysiology, symptoms, physical examination, further examination, differential diagnosis, and therapy. Thinking-aloud, explanation, and the learner reports were audio-recorded and transcribed for further analysis.

Analysis

Progress of knowledge use

The progress of knowledge use over the cases up to the learner reports was charted by identifying specific concepts in the protocols. These concepts were based on a list of relevant encapsulating concepts for the case series, which together explain the fault in the cases (see Appendix 1). Similar to the examples given above, these concepts differ in depth and explanatory power. Two concepts in both series are directly linked to observable consequences; those concepts are termed superficial here. Another two concepts in both series are also directly linked to consequences but are less observable. They are termed intermediate here. The rest of the lists consist of concepts describing biomedical processes and principles that are indirectly linked to enabling conditions and consequences. They are described as deep (biomedical) concepts. The PCO cases are explained by nine deep concepts, the HPA cases by ten. These lists were developed in cooperation with four domain experts. Furthermore, development of misconceptions, reminders of earlier cases, and language use (professional versus everyday language) were investigated.

Elaborate reasoning about the cases was seen during case processing (14 participants) and afterward during the pathophysiological explanation (12 participants), whilst 4 of them did so in both tasks. Therefore, these documents were analysed in combination. Outcomes were compared with the learner reports.

Learning from the cases

The analysis of the learning process as expressed in the think-aloud protocols, the pathophysiological explanations, and the learner reports were developed in an iterative way. Coding categories and scoring criteria were developed by going through a subset of cases repeatedly until no changes were needed between cycles. The coding categories regarded the phase in the learning process where most thinking took place (thinking aloud, explaining the case, or in the learner report), occurrence of reminders of earlier cases, biomedical reasoning, improvement over cases, remediation of misconceptions over cases as well as in the learning reports, and language use. Interrater agreement over another subset of 10 % revealed an average agreement of 73 %.

Content analysis of the learner reports

The third analysis regarded the results of knowledge development as expressed in the learner reports. We took a model text description of that disorder, derived from medical textbooks and checked by two domain experts as a basis for comparison. The model text was segmented into meaningful wholes and classified into the same categories of clinical

texts that were used as probes. The segments were complex statements consisting of several propositions. An example is 'affected are females in the fertile stage of life'. The numbers of segments per category were identical in the two cases. The following categories were added, as the learner reports happened to include statements that did not fit the standard categories: normal physiology (generic knowledge), inferences, and elaborations. Utterances were mapped on these segments and coded as complete or partial fits. Utterances were classified as inferences when they could be described as conclusions students drew within the relevant knowledge domain; they were classified as elaborations when they were additions based on other related knowledge fields. Furthermore, the number of metacognitive statements referring to knowledge monitoring processes (e.g., 'I don't know if …') were counted. Inferences, elaborations, and reasoning based on normal physiology were categorised separately and taken as indicators for active knowledge construction and reconstruction. Definition had three segments, epidemiology two, pathophysiology 21, symptoms 14, physical examination six, further examination five, differential diagnosis three, and therapy had seven segments. Scores in the pathophysiology category represent the use of problem-oriented knowledge, the faults in illness scripts. Epidemiology represents the illness script category enabling conditions; symptoms and (additional) physical examination regard consequences. Two independent raters coded the first 10 learner reports. The raters compared and discussed differences until they came to a full agreement. The remaining learner reports were coded by one rater. Quantitative learner report measures were analysed with a multivariate analysis of variance (MANOVA) with repeated measures.

Results

Progress of knowledge use

Most students used more explicit, detailed reasoning in the first case than in later cases in the series. This pattern was more pronounced in the PCO group. Many students diligently activated and sometimes painfully reconstructed their knowledge (e.g., participant 436, see Table A in Online Appendix) when dealing with the first case.

Use of specific fault concepts

The analyses of specific fault concepts (see Appendix 1) that were used in diagnosing and explaining the cases and in the learner reports revealed that relatively most of the fault concepts the students used were superficial, followed by the intermediate concepts, and then followed by the deep concepts (Table 1). Three values stood out in this pattern: the relatively high number and proportion of deep concepts in the HPA protocols, the overall small proportion of intermediate concepts in the HPA learner reports, and the high proportion of intermediate concepts in the PCO protocols. No general trend was discernable in the emergence or disappearance of deep concepts. In the intermediate concepts we saw, however, that the concept of 'endometrial proliferation', which was most prominent in the first case presented, was followed by 'anovulation' in the later cases of the series, indicating that the students extended their case representation over multiple cases. The most explicit examples are participants 415, 416, and 428 (see Table C in Online Appendix).

Table 1 Total number of relevant encapsulating concepts per case series in diagnosing and explaining the cases (think aloud and pathofysiological explanation protocols) and in the learner reports for each of the two case series groups (weighed mean use per concept per group and participant in brackets)

	HPA $N = 14$		PCO $N = 17$	
	Diagnosis and explanation	Learner report	Diagnosis and explanation	Learner report
Superficial[a]	26 (.93)	15 (.54)	34 (1)	31 (.91)
Intermediate[a]	11 (.39)	2 (.07)	23 (.68)	11 (.32)
Deep[b]	46 (.33)	17 (.12)	31 (.20)	17 (.11)

[a] 2 concepts

[b] 10 and 9 concepts, respectively

Learning from the cases

Of the 31 participants, only 15 demonstrated in their dealings with the cases that they learned from solving the cases. These participants became less uncertain, used more specific language, gradually came to focus on the essential aspects of the biomedical process, or showed more complete pathophysiological explanations (see Tables B and D in Online Appendix). The following examples illustrate what students learned. One student evaluated his prior knowledge in light of the cases dealt with: "It concerns a certain hormonal disorder, in which I thought to have *learned from literature* that LH *is always* high, oestrogen high and FSH low … *but the cases show* that it is sometimes different" (participant 422). Another concluded in the learner report that she had learned something *new*: "I did not know it were so many [follicles]" (p 413). The other 16 students may have learned but did not show this in their approach to the case, or in their learning reports.

Learning from multiple cases can be assumed to be most effective when students have well-developed validated knowledge. The think-aloud protocols, explanations, and learner reports showed that the knowledge of the majority of the participants in this study did not really meet this criterion (Table 2 and see Tables A and C in Online Appendix). No participant had all of the relevant knowledge available; 7 had very poor prior knowledge. The other 24 had basic relevant knowledge that was, however, incomplete and error-prone. Fourteen engaged in active biomedical reasoning aimed at improving errors or uncertainties the students had recognised. This active debugging in the HPA series was mostly done within a single case; in the PCO series, active debugging was extended over cases. Six participants appeared able to—partially—remediate the misconceptions they had expressed, whereas 2 participants chose to further ignore the errors they noticed. The remaining 14 participants were either not aware that they made errors of reasoning or were incapable of solving them, as the participants made errors persistently throughout the case series. The most common and persistent misconception was that excessive amounts of oestrogens are converted into androgens in fat cells leading to an androgenic appearance in affected females, whilst it happens to be the other way round.

Most misconceptions regarded the interactions between the different hormones ($n = 18$). Most prominent in the HPA cases were lack of knowledge ($n = 8$) and misconceptions about the production or the role of GnRH (a releasing hormone produced in the hypothalamus) ($n = 3$). In the PCO cases, the participants who mentioned peripheral hormone conversion all showed clear misconceptions about the process ($n = 5$). Mostly

Table 2 Quality of the students' knowledge, ways of reasoning and dealing with errors as indicated by the number of participants per category

	HPA[a] $N = 14$	PCO[b] $N = 17$
Poor prior knowledge	3	4
Incomplete and error-prone knowledge	11	13
Active reasoning, aimed at improvement	5	9
Remediation of errors	3	3
Ignores noticed error	–	2
Persistent errors	8	5

such expressions were accompanied by expressions of doubt. Misconceptions were never rectified in the learner reports; the misconceptions were merely left out of the report.

Many students ($n = 25$) stated in their think-aloud protocols that they had been reminded of a previous case in the series. Sometimes the students explicitly compared the findings. For instance, participant 431 started her learner report with "From the different cases I have seen how HPA can manifest itself", whilst participant 413 explicitly compared the size of the ovaries and numbers of cysts in the different cases. Some students distrusted the similarities due to the artificial situation. Half of the students ($n = 15$) stated in their learner reports that they had learned from the variation in the cases. Several participants in the PCO group ($n = 6$ out of 8 who referred to the cases) emphasised the variation in symptoms that had struck them. Half of the participants in the HPA group ($n = 7$) also attended to the variations in pathophysiology. One student explicitly reported on what she had incorrectly thought before (that "there was no ovulation because of a lacking production of oestrogens") but now knew that there can be a "rather high oestrogen level and yet no ovulation" (participant 412). No students clearly expressed that they had learned from the reasoning process itself, although one student stated after he had explained the third case: "Two more PCO cases and I know the clinical picture of PCO again" (participant 422).

Knowledge used in the learner reports

The outcomes of the content analysis of the learner reports show that the statements about illness script categories mainly regard the consequences (symptoms and additional physical examination) and the fault (pathophysiology). The number of enabling conditions statements is low.

Results for the learner reports showed that the students who had worked on the PCO series were more extensive in their reports by using more utterances than the students who had worked on the HPA cases [$F(1,29) = 15.12$, MSe $= 2.17$; $p = .001$, partial $\eta^2 = .34$, $OP = .96$; means 13.21 for the PCO group and 6.42 for the HPA group].

The significant interaction effect [$F(4.853,140.747) = 4.57$; MSe $= 2.12$, $p = .001$, partial $\eta^2 = .14$, $OP = .97$] was especially due to the categories (see Fig. 2) normal physiology, inferences and elaborations (PCO > HPA), indicating that the group who had worked on the PCO series put more reasoning effort into biomedical reasoning to fill gaps in their fault knowledge. The fact that the HPA group scored higher on pathophysiology suggested that this group's knowledge better fitted the cases presented. The pathophysiology category had the highest overall mean. Furthermore, the students in the PCO group tended to generate more metacognitive utterances (5.24 vs. 3.64), but this difference was not significant. Of the four categories that were taken as indicators for illness

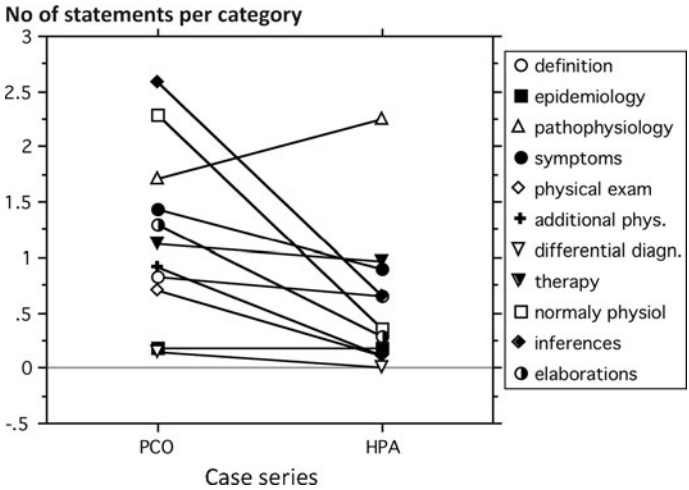

Fig. 2 Elements of learner reports of students who had either worked on the PCO series or on the HPA series

script categories (epidemiology, symptoms, physical examination, and additional physical examination), only the latter two showed significant differences ($p < .05$; PCO > HPA).

Discussion

This study departed from the observation that research and theory on illness script construction by medical students emphasises the importance of the integration of—separately taught—biomedical fault knowledge. This process requires restructuring of biomedical knowledge, which is assumed to take place as a function of the application of such knowledge on clinical problems. Within the limits of preclinical education, these processes are supervised, for instance, in tutorial groups in PBL, but in clinical teaching settings, teachers focus different things, e.g., differential diagnosis and treatment, whilst the students' main task is to take patients' full history and examination and report that in the patients' records. Whilst in preclinical settings, the curriculum and teachers define which cases are presented; in the clinical settings, it is the epidemiology of the time of the year and specific hospital or outpatient clinic. Furthermore, feedback on the results is often delayed, if indeed the doctor or student ever receives that feedback. This makes the clinic a reduced learning setting. In this context, we situated our concern about what medical students learn spontaneously from reasoning through a series of cases that have a similar underlying biomedical problem and where only reduced feedback is provided. More specifically, we wanted to know how knowledge about the constituent parts of illness scripts (enabling conditions, fault, and consequences) develop, where we left the possibility open that different parts do not develop in the same way.

The results can be summarised as follows:

(1) Students did learn from processing multiple cases even when the feedback is restricted to the diagnosis. They built up representations faster, became more secure,

used more specific language, and tested their knowledge against case information. Yet half of the students did not show this kind of learning.

(2) Most students learned from the variations in enabling conditions and consequences components of the cases. Some students compared prior knowledge with case findings, some actively compared cases, drawing conclusions about enabling conditions and consequences, whilst others seemed to learn implicitly. No student made errors in their conclusions.

(3) Learning from a series of cases appeared to have less positive outcomes regarding the fault component of illness scripts. Many students had poor knowledge, including misconceptions. Some tried, and one even succeeded in debugging these misconceptions, but the results of these enterprises never made it to the learner reports. Since one of the probes for the learner report invited students to report on what they knew now about the pathophysiology of the disease, this finding seems to indicate that the students' knowledge did not really improve. Other indicators for this lack of learning or improving problem-oriented knowledge were (a) relative lack of use of deep encapsulating concepts especially regarding endocrine regulation and (b) application of generic biomedical knowledge and the emergence of elaborations and inferences in the learner reports. These findings suggest that knowledge construction and reconstruction were still in progress and were hampered by a lack of knowledge and by deep misconceptions.

In summary, we may conclude that there is much *implicit* learning, which may lead to faster recognition, strengthening of the links between concepts, and tuning of knowledge to practical conditions (Anderson 1982). As many students did not show any explicit learning processes, this result raises doubts about the students' ability to take full advantage of learning from cases. Furthermore, *explicit* learning anchored by given case findings consisted of two types: comparison of case findings with prior knowledge and explicit comparison of cases. Both processes had positive outcomes, but in the current setting, this learning is very dependent on whether the students' attention is also drawn to co- and invariation between cases (Marton and Pang 2006). Finally, explicit case explanations based on deep concepts resulted in knowledge construction and debugging processes that are considered to be conducive to knowledge network validation and (re)structuring, but students were unable to complete these processes unsupported. We may conclude from these analyses that students may learn patterns of and variations in enabling conditions and consequences from this kind of learning situation but that unsupported case analyses are in no way sufficient for integrating biomedical knowledge in illness scripts.

Before answering the question which learning processes can be left to their own devices and what the implications for teaching are, we must mention that the teaching hospital, despite the term, is primarily a workplace where patients come first, and where medical students work and learn. Yet in this environment, patients are the first learning resources, apart from the medical and nursing staff and reference materials such as books and databases (Dornan et al. 2008). In this context, we tend to conclude that students are well able to learn the patterns and variations of enabling conditions and consequences independently. However, learning can become more fruitful if students are supported to build a case-base they can reflect on in a systematic way. Modern technology such as personal digital assistants become rapidly integrated in student learning in the clinic (Garrett and Jackson 2006; Ranson et al. 2007; Tempelhof 2009) as a resource or as an access point to the students' portfolio where they can upload case information and reflections. Buckley et al. (2009) concluded in their review study that the higher-quality studies found suggested

that learning portfolios help medical students improve the integration of theory with practice, and lead to greater self-awareness and reflection. Integration with the hospital information and communication technologies-systems might be very worthwhile for learning but is hindered by system safety measures and by patient privacy considerations.

The present results do not tell whether students will use resource information to debug a misconception in biomedical knowledge, which is a required condition for integration. As one—both teacher and student—is aware of poor knowledge and misconceptions only during problem-solving, and since debugging is an effortful enterprise, it is doubtful whether opportunities for impromptu case-based teaching events in the clinical practice (e.g., in outpatient clinics, or grand rounds) are sufficient. On top of that, active knowledge debugging and integration should also take place off-line, outside the situation where the problem was detected. Kassirer (2010) suggested a potentially powerful case conference method, in which a maximum of 30 students, clerks, or interns participate. Participants bring in their own cases. This is where the method might integrate with portfolio work. A coach who is unfamiliar with the cases best supports the group in case analysis. The main goal of these conferences is optimising clinical reasoning, but an important rule in the reasoning process is "Working diagnoses, that is, hypotheses used for prognostic or therapeutic recommendations, are evoked only after they are assessed for their adequacy in explaining all positive, negative, and normal clinical findings, and for their pathophysiologic reliability—namely, a check on the reasonableness of causal linkages between clinical events" (Kassirer 2010, p. 1119). The role of the coach is that all participants are active and to prevent cognitive errors from being made. Though Kassirer himself focused on clinical reasoning skill, cognitive restructuring should be a by-product of the process.

The focus of this study was theory development, that is, to find out what students can learn spontaneously from multiple cases. The outcomes were to a large extent influenced by the specific case mixes. If medical educators were right that the topics at hand were important and relatively easy, it is hard to say what will happen with more difficult topics. The instructional conclusions based on these outcomes regarded the clinical period in which further learning should take place. Findings at the point of transition from one learning environment to the next, or from theory to practice, always generate the question whether education in the previous period should not have provided for this problem. Apart from a policy issue, this issue also concerns pedagogical content knowledge. Prevention of misconceptions is not a big issue in medical education literature. Feltovich et al. (1997) warned against simplification in such complex domains and instead advised multiple perspectives and metaphors to make students appreciate the full complexity of the biomedical domain.

The strength of the present study is that it investigated unsupported, spontaneous learning, which allowed us to draw conclusions about the learning processes and outcomes. Some of these conclusions were based on what was present in the protocols; others were based on what was not present. Though the latter conclusions play a small role in the implications formulated, especially conclusions about implicit learning and not learning should be further investigated with other methods.

Appendix

See Table 3.

Table 3 Relevant fault concepts for the HPA and PCO cases

HPA	PCO
Superficial concepts	Superficial concepts
Galactorrhea	Hirsutisme
Irregular cycles with withdrawal bleedings	Irregular cycles with withdrawal bleedings
Intermediate concepts	Intermediate concepts
Endometrium proliferation	Endometrium proliferation
Anovulation	Anovulation
Deep concepts	Deep concepts
Hypothalamic-pituitary-end organ axes	Hypothalamic-pituitary-end organ axes
Compression (damage)	Follicles; functional - hormonal secretion and receptors
Hypothyrodism	Hyper-oestrogenic environment
Hypogonadism	Hyper-androgenic (ovaries)
Dopamine antagonist	Vicious circle
Negative feedback of prolactin on GnRH	Feedback oestrogens on LH and FSH
Negative feedback of T3 and T4 on TSH	Peripheral conversion of andosteendiol into oestrogen
Disconnection	Inhibition
Inhibition	Stimulation
Stimulation	

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